Right at the beginning of *Six Easy Pieces* we learn how all physics is rooted in the notion of law—the existence of an ordered universe that can be understood by the application of rational reasoning. However, the laws of physics are not transparent to us in our direct observations of nature. They are frustratingly hidden, subtly encoded in the phenomena we study. The arcane procedures of the physicist—a mixture of carefully designed experimentation and mathematical theorizing—are needed to unveil the underlying law-like reality.

Possibly the best-known law of physics is Newton's inverse square law of gravitation, discussed here in Chapter Five. The topic is introduced in the context of the solar system and Kepler's laws of planetary motion. But gravitation is universal, applying across the cosmos, enabling Feynman to spice his account with examples from astronomy and cosmology. Commenting on a picture of a globular cluster somehow held together by unseen forces, he waxes lyrical: "If one cannot see gravitation acting here, he has no soul."

Other laws are known that refer to the various nongravitational forces of nature that describe how particles of matter interact with each other. There is but a handful of these forces, and Feynman himself holds the considerable distinction of being one of the few scientists in history to discover a new law of physics, pertaining to the way that a weak nuclear force affects the behavior of certain subatomic particles.

High-energy particle physics was the jewel in the crown of postwar science, at once awesome and glamorous, with its huge accelerator machines and seemingly unending list of newly discovered subatomic particles. Feynman's research was directed mostly toward making sense of the results of this enterprise. A great unifying theme among particle physicists has been the role of symmetry and conservation laws in bringing order to the subatomic zoo.

As it happens, many of the symmetries known to particle physicists were familiar already in classical physics. Chief among these are the symmetries that arise from the homogeneity of space and time. Take time: apart from cosmology, where the big bang marked the beginning of time, there is nothing in physics to distinguish one moment of time from the next. Physicists say that the world is "invariant under time translations," meaning that whether you take midnight or midday to be the zero of time in your measurements, it makes no difference to the description of physical phenomena. Physical processes do not depend on an absolute zero of time. It turns out that this symmetry under time translation directly implies one of the most basic, and also most useful, laws of physics: the law of conservation of energy. This law